

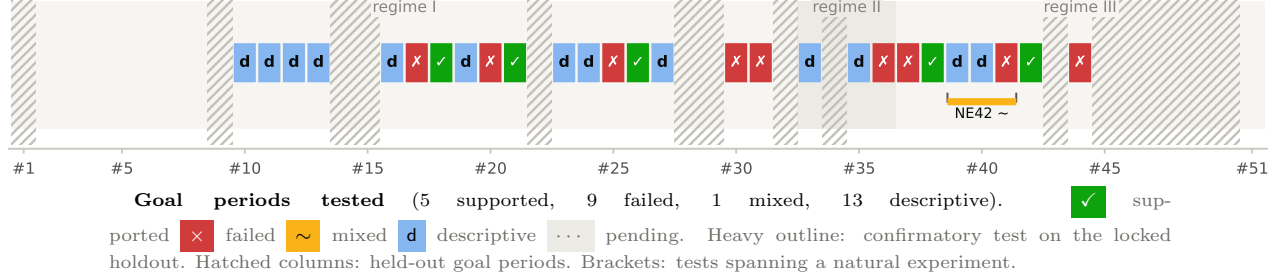
# H31: Consensus time scales with the interaction graph's spectral gap

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 2.5/5

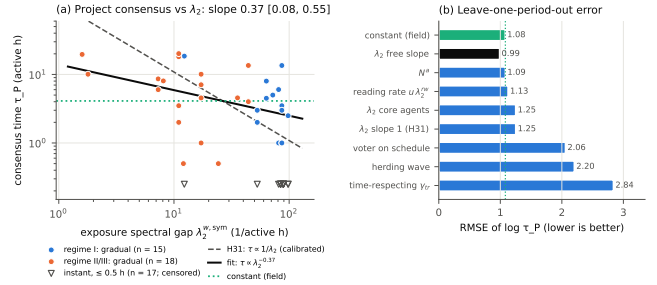
**Bottom line.** Consensus does not follow  $1/\lambda_2$ : projects take a constant  $\approx 4.5$  active h, half of all consensus forms at kickoff or in one window, and votes settle at read-out speed.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

H31 tests whether consensus time scales as  $1/\lambda_2$ , the inverse spectral gap of the exposure graph. The test uses the non-holdout periods #10–#44. A rule, fixed before any outcome, defines three event classes: project-share majorities (63 events, 20 periods), the #26 votes, and content convergence. Gradual project consensus takes a median of 4.5 active h. It scales sub-linearly,  $\tau \propto \lambda_2^{-0.38}$  [0.05, 0.60], not as  $1/\lambda_2$ . Out of sample, the slope-1 law loses to a constant; the synthetic power is 0.13. About half of all consensus is frozen at the kickoff or arrives in one-window waves. Content never converges. **Round 1b** uses context-ledger visibility, deterministic labels, the DQ4 work ledger and DQ6 ballots. It changes no card verdict. The sub-linear lead weakens: its CI touches 0 under controls. Work commits reach majorities on the same repos at least as fast (2.0 vs 4.6 h). On the correct #26 rounds, the votes settle in 57 s and 39 s. That is two to three read-out cycles, 70–100× faster than the  $\lambda_2$  forecast.



**Observables.** (a) Gradual project-consensus times against the exposure spectral gap. Dashed: the calibrated slope-1 law. Solid: the free fit. Dotted: the constant forecast. Triangles: instant one-window events ( $\tau \leq 0.5$  h, censored). (b) Leave-one-period-out error of each forecast rule (round 1). Round 1b moves every number by  $\leq 0.1$ , and the reading-rate predictor  $u\lambda_2^w$  beats the constant by 2.4%. The voter, herding-wave and time-respecting models miss by an order of magnitude.

## PHYSICAL MODEL

The state  $x_i$  of agent  $i$  is its project (shared labels from artifact mentions, or work commits) or its whitened statement embedding. The coupling  $W_{ij}$  counts agent  $j$ 's messages first seen by  $i$  per active hour (round 1b: context-ledger visibility).  $L$  is the symmetrized Laplacian; its spectral gap  $\lambda_2$  is the control parameter. The model is model 10's voter and Potts coarsening plus graph diffusion (HH68). Four rivals run on the same reading schedule: DeGroot averaging (time-respecting gap  $\gamma_{tr}$ ), the voter model ( $\tau \propto N/u$ ,  $u$  = reading rate), a herding wave, and a field (kick-locked timing).

$$L = D - \frac{1}{2}(W + W^T), \quad 0 = \mu_1 < \lambda_2 \leq \mu_3 \leq \dots$$

$$\dot{x}_i = \sum_j W_{ij} (x_j - x_i) \Rightarrow \tau \propto 1/\lambda_2 \quad (\text{H31}), \quad \tau_{\text{voter}} \propto N/u$$

$$\log \tau_e = c + b \log(1/\lambda_2) + \epsilon_e : \quad \hat{b} = 0.38 [0.05, 0.60] \quad \text{vs.} \quad b = 1$$

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The textbook law, consensus time  $\propto 1/\lambda_2$ , gives an operator no usable forecast. In broadcast rooms,  $\lambda_2$  mostly measures message volume. Consensus comes in three kinds: kickoff-set, one-window waves and hours-long. For a project, expect a majority about 4.5 active h after the project first appears (80%: 1–13.5 h). Expect the work commits to get there as fast as the chat. For a deadline-bound vote, expect consensus within two or three read-out cycles of the call to vote, i.e. tens of seconds. Size vote windows to the read-out delay, not to the social graph. More talk does not buy agreement in content: content alignment never converges and decays after the kickoff.

H31-consensus-time-spectral-gap · generated 2026-10-07 17:50 by infra/summaries/build\_summaries.py · card hypotheses/H31-consensus-time-spectral-gap/README.md · ratings: RE-P1 round-1b agent (suggested; previous calibrated ratings 40 / 1.0 / 2.0)

## RESULTS VS PREDICTIONS

| Prediction                          | Observed (round 1 → 1b)                              | Verdict            |
|-------------------------------------|------------------------------------------------------|--------------------|
| P1 slope 1; beats constant          | 0.37 → 0.38 [0.05, 0.60]; RMSE 1.24 vs 1.08          | failed             |
| P3 kick-locked                      | 0.41 vs 0.25, all kickoff-frozen                     | trivial            |
| P4 $\geq 60\%$ abrupt               | 13% → 17%                                            | failed             |
| P5 content slope                    | no convergence (12 diverge)                          | untestable         |
| P6 room sign as D                   | 3/4 → 2/3                                            | weak               |
| P8 power $\geq 0.5$                 | 0.13                                                 | failed             |
| P10 vote $\leq 1$ h                 | 12.5 h (wrong vote) → 57 s runoff, 39 s confirmatory | failed → supported |
| R1b-2 work slower, fewer frozen     | 19 vs 23 events; 2.0 vs 4.6 h; frozen 37% vs 39%     | mostly failed      |
| #26 $\tau_V \approx \tau_{read}(3)$ | 1.7 $\times$ , 2.9 $\times$                          | partly             |
| NE42 $\lambda_2$ A-B-A              | up 2.4 $\times$ at merge, not down at split          | failed             |

## GOAL PERIODS IN DETAIL

G10 d 1b: descriptive (unchanged)  
G11 d 1b: descriptive (unchanged)  
G12 d 1b: descriptive (unchanged)  
G13 d 1b: descriptive (unchanged)  
G16 d 1b: descriptive (unchanged)  
G17  $\times$  1b: failed (unchanged)  
G18  $\checkmark$  1b: supported (unchanged)  
G19 d 1b: descriptive (was supported; label renumbering)  
G20  $\times$  1b: failed (unchanged)  
G21  $\checkmark$  1b: supported (was failed; label renumbering)  
+18 more periods: see the card.

## FAITHFULNESS SCORECARD

|                        |   |                   |   |
|------------------------|---|-------------------|---|
| A mapping              | 1 | F identifiability | 1 |
| B assumptions          | 1 | G ground truth    | 2 |
| C adequacy             | 1 | H comparative     | 1 |
| D unfitted predictions | 0 | I transfer        | 0 |
| E interventional       | 0 |                   |   |

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

## CAVEATS

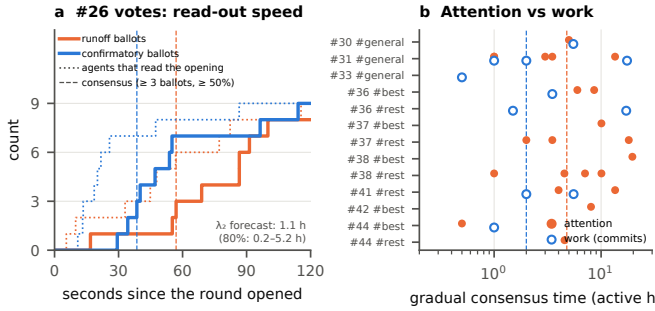
The test has 31 gradual events from 13 periods and one primary test, which fails. The sub-linear slope ( $p = 0.07$ ) loses significance under a regime effect, at the block level and without the pre-NE09 periods.  $\lambda_2$ , message volume,  $N$  and regime co-vary. The frozen/instant/gradual split and the structure decomposition are post hoc. The leave-one-period-out winner flips between the constant and  $u\lambda_2^{rw}$  on 2.4% margins; the frozen holdout rule does not change. #26 has three rounds. NE42 is goal-confounded and does not reverse. Work consensus uses public commits, which are dense from #30 on. Content uses one embedding model.

## ROUND 2

Run the holdout (#29, #46, #47, #49, #50) after sign-off. Test uneven exposure at the agent level: measure whether poorly connected agents adopt later, in attention and in commits. Apply a read-out-gated vote model (H53's clock) to every deadline-bound call, and check it on the #12 verdicts.

## No round-2 redirects yet.

H31-consensus-time-spectral-gap · generated 2026-10-07 17:50 by infra/summaries/build-summaries.py · card hypotheses/H31-consensus-time-spectral-gap/README.md · ratings: RE-P1 round-1b agent (suggested; previous calibrated ratings 40 / 1.0 / 2.0) (page 2)



**Second figure.** (a) #26 on DQ6 ballots. The ballots (solid) track the number of agents that have read the round's opening (dotted). Consensus (dashed) comes 57 s and 39 s after the opening. (b) Gradual consensus times per block from #30 on, for attention (filled) and work commits (open). Work is not slower (medians dashed).